

Large Language Models (LLMs) and Their Financial Application

Research Project

Bagus Pranata

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WORD COUNT: 7,241

List of Abbreviations

BERT	<i>Bidirectional Encoder Representations from Transformers</i>
BPS	<i>Basis points (in context of stock' transaction cost)</i>
CRSP	<i>Center for Research in Security Prices</i>
DistilBERT	<i>Distilled BERT</i>
DistilRoBERTa	<i>Distilled RoBERTa</i>
EW	<i>Equal-Weighted</i>
FinBERT	<i>Financial BERT</i>
GPU	<i>Graphics Processing Unit</i>
IQ	<i>IQ Key Development</i>
L	<i>Long strategy</i>
LLMs	<i>Large Language Models</i>
LS	<i>Long-Short strategy</i>
RNN	<i>Recurrent Neural Network</i>
RoBERTa	<i>Robustly optimized BERT approach</i>
S	<i>Short strategy</i>
SR	<i>Sharpe Ratio</i>
US	<i>United States</i>
VW	<i>Value-Weighted</i>
WRDS	<i>Wharton Research Data Services</i>

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Abstract

This study investigates the application of Large Language Models (LLMs) to predict stock returns to build portfolios in the US Communication Service Industry sector by combining sophisticated LLMs (such as BERT, RoBERTa, DistilBERT, DistilRoBERTa, and FinBERT) with standard econometric approaches. By using a comprehensive method that includes collecting and combining data, analysing sentiment, and doing rolling window analysis, the models were optimised to more accurately reflect market perceptions and patterns. The findings indicate that fine-tuning greatly enhances the capabilities of the models, with BERT and DistilBERT exhibiting greater accuracy and risk-adjusted returns. The efficiency of these models in balancing risk and return in long-short portfolios is highlighted by detailed performance assessments, which include accuracy measures and Sharpe Ratios. The results emphasise the significance of adapting LLMs for financial purposes, indicating significant implications for portfolio management, transaction costs, and wider financial analysis. Although the research shows positive results, it also recognizes limits such as variations in performance across different models and the need for more extensive datasets. It is suggested that future study should focus on improving LLMs, investigating more sophisticated fine-tuning methods, and dealing with ethical and regulatory issues in algorithmic trading.

Acknowledgements

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Declaration

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Chapter 1: Introduction

1.1 Background & Problem Statement

Machine learning and natural language processing have had a significant impact on the financial industry, leading to profound transformations. Large Language Models (LLMs) have shown remarkable proficiency in processing and analysing textual data. These models possess the ability to comprehend complex patterns in language, rendering them significantly valuable for forecasting market fluctuations by analysing news and sentiment.

The US Communication Service Industry, comprising major corporations such as Google (Alphabet), Netflix, Walt Disney, and AT&T, is extremely vulnerable to market sentiment and news events. Precise forecasting of stock returns in this industry can provide substantial advantages to investors and stakeholders. Nevertheless, the task of incorporating and exploiting high-frequency textual data for financial forecasts continues to be a challenge.

This study aims to tackle this challenge by examining the use of LLMs to perform sentiment analysis on headline news. The study further aims to use this analysis to predict stock returns and construct portfolios specifically for the US Communication Service Industry. To access a comprehensive list of the companies that used in this study and their respective market capitalizations, please refer to the Appendix A.

1.2 Shape and Scope of the Study

This study addresses several broad questions:

- What is the impact of extracting specific sentiment, such as positivity or negativity, from headline news on predictions of stock returns?
- What is the comparative efficacy of full-model versus distilled LLMs in processing and analysing financial news?
- Does incorporating sentiment data into stock return data improve the accuracy of predictive models designed for portfolio construction and risk assessment?

The scope includes Top-25 communication service companies in the United States, and the data covers a substantial period of time to ensure the analysis is robust and reliable. The focus is on developing a predictive model that integrates LLM-based sentiment analysis with conventional econometric techniques in order to predict stock returns and build into a portfolio.

1.3 Structure of the Report

The report is structured as follows:

- **Chapter 1: Introduction** This chapter introduces the background and problem statement, study's shape and scope, and presents the overall structure of the report.
- **Chapter 2: Literature Review** This chapter provides an overview of the existing research conducted on LLMs and their specific applications in the field of financial forecasting and its transaction costs. It includes a comprehensive overview of LLMs, encompassing their models, architecture, and their use case in financial forecasting.
- **Chapter 3: Methodology** This chapter provides an explanation of the study design and methodology employed in the study. It covers the conceptual framework, specifications of the hardware, data collection and preprocessing, analysis using rolling windows, sentiment analysis, and the techniques employed for forecasting stock returns and building the portfolio.
- **Chapter 4: Results and Discussion** This chapter showcases the practical findings of the study, assesses the effectiveness of the predictive models, and explores the significance of the results to the portfolio.
- **Chapter 5: Conclusion & Future Work** This chapter concludes and summarises the key findings, examines the consequences for both theory and practice, acknowledges the study's limitations, and provides recommendations for future research.

Chapter 2: Literature Review

2.1 Overview of Large Language Models

Large Language Models (LLMs) have transformed the field of natural language processing (NLP) by offering advanced capabilities in understanding and generating human language. The concept of LLMs was first introduced with the development of Eliza in the 1960s, which was the pioneering chatbot in the world. Eliza initiated the exploration of NLP, establishing the groundwork for subsequent, more intricate LLMs (Weizenbaum, 1966). LLMs, or Language Models, are more successful than conventional word-based approaches because they take into account the syntactic and semantic intricacies of language. This makes them well-suited for tasks like sentiment analysis, machine translation, until text summarization. (Devlin et al., 2018; Radford et al., 2019).

The emergence of LLMs started with the introduction of models such as Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014). These models portrayed words as unchanging vectors in a space with several dimensions. The initial models served as the foundation for more advanced architectures such as BERT and GPT . These architectures employ transformer networks to process text in both directions and generate output, respectively (Devlin et al., 2018; Radford et al., 2019).

2.2 LLM Models and Architecture

LLMs often use transformer networks, which have encoders and decoders arranged in many layers. Vaswani et al. (2017) created Transformers, which represented a notable advancement in LLMs by surpassing earlier models such as Recurrent Neural Networks (RNNs) in their capacity to properly handle long-range dependencies in text. Transformers, in contrast to RNNs, use self-attention mechanisms to concurrently analyse all tokens in a sequence, enabling improved parallelization and comprehension of context (Vaswani et al., 2017). Table 1 presents a comparison of several LLM architecture and their respective characteristics used in this study:

Table 1. Models architecture and characteristics comparison

Model	Number of Parameters	Pre-training Data	Architecture and Characteristic
BERT	110M	BooksCorpus, Wikipedia	Bidirectional, Masked Language Modelling
RoBERTa	125M	BooksCorpus, Wikipedia	Enhanced training procedure, larger batch sizes
FinBERT	85M	Financial News Corpus, FinBERT	Domain-specific pre-training for financial texts
Distil BERT	66M	BooksCorpus, Wikipedia	Distilled version of BERT, smaller and faster
Distil RoBERTa	82M	BooksCorpus, Wikipedia	Distilled version of RoBERTa, focused on efficiency

Note: This table compares different LLMs based on their number of parameters, pre-training data sources, and key architectural characteristics, highlighting the strengths and specialised features of each model.

BERT, a model created by Devlin et al. (2018), employs a bidirectional strategy for masked language modelling, enabling it to comprehend the contextual meaning of a word in both forward and backward directions. The bidirectional nature is essential for understanding intricate language patterns and nuances. BERT undergoes pre-training using the BooksCorpus and Wikipedia datasets, which include a total of 110 million parameters. This extensive pre-training equips BERT with the ability to comprehend and incorporate a wide variety of linguistic elements and contextual information.

RoBERTa, a model developed by Liu et al. (2019), improves upon BERT by including bigger batch sizes and more training data into the training process. The objective of this technique is to enhance the resilience and precision of the model. RoBERTa, which has been pre-trained on BooksCorpus and Wikipedia, consists of 125 million parameters, making it one of the largest and most powerful models in terms of parameter size.

FinBERT is a specialised model designed specifically for analysing financial language. FinBERT, created by Yang et al. (2019), utilises a financial news corpus and FinBERT data during pre-training to enhance its ability to comprehend and analyse financial language and terms with more efficiency. FinBERT, which has 85 million parameters, is specifically designed for jobs involving financial sentiment analysis and forecasting.

DistilBERT, a condensed iteration of BERT, was developed by Sanh et al. (2019) with the objective of reducing size and enhancing speed, while still maintaining a significant portion of BERT's precision. DistilBERT, which has 66 million parameters, undergoes pre-training using the same datasets as BERT. Similarly, DistilRoBERTa, created by Sanh et al. (2019), is

a condensed iteration of RoBERTa. The major objective is to enhance efficiency without compromising performance, using a total of 82 million parameters.

2.3 LLM Application in Stock Forecasting

The use of LLMs in stock forecasting is an expanding field of study. Conventional methods of financial prediction often depend on numerical data and technological indicators. Nevertheless, including textual data, such as news articles, earnings reports, and social media postings, might provide supplementary information that can improve prediction models (Bybee et al., 2020; Gu et al., 2020).



Figure 1. Flowchart of key studies and findings related to LLMs in Financial Forecasting.

Multiple studies have shown that LLMs are successful at extracting sentiment from text data and applying it to forecast stock returns. Figure 1 provides a concise overview of key studies and findings about the use of LLMs in financial prediction. Chen et al. (2022) demonstrated that using contextualised representations of news content via LLMs yields superior results compared to conventional word-based approaches for forecasting stock returns. Their research included data from several global stock markets and emphasised the capacity of LLMs to collect intricate information that impacts market sentiment and consequent stock prices.

LLMs provide a more advanced analysis of text, taking into account elements like negation, sarcasm, and complex syntactic patterns that less complicated models may overlook. Wu et al. (2020) investigated the use of LLMs as few-shot learners for financial sentiment research, while Bybee et al. (2020) examined the structural attributes of economic news using LLMs. Conneau et al. (2020) expanded on this by investigating cross-lingual representation learning,

which has implications for financial sentiment analysis. As a result, there is an improvement in sentiment analysis accuracy and enhanced predictive capabilities in stock forecasting models (Devlin et al., 2018; Liu et al., 2019).

2.4 Transaction Costs in Financial Trading

Transaction costs are an essential factor to consider when analysing and implementing financial trading models. Frazzini et al. (2013) conduct a thorough analysis of transaction costs and observe that the median NYSE market equity serves as the size threshold for the U.S. sample. Stocks that exceed this threshold are categorised as large-cap, and those that fall below it are regarded as small-cap. It emphasises that the typical trade of a large-cap company results in market impact fees of 10.25 basis points, while small-cap equities suffer 21 basis points. The impact of these expenses on net profits and risk-adjusted performance is substantial, underscoring the necessity of integrating transaction costs into financial models to achieve accurate and practical insights.

Chapter 3: Methodology

3.1 Conceptual Framework

The study's conceptual framework combines LLMs with conventional econometric techniques, such as logistic regression, to forecast stock returns and conduct sentiment analysis specifically for the US Communication Service Industry. The purpose of this integration is to utilise the sophisticated capabilities of LLMs in processing and interpreting textual data, while also leveraging the strength and reliability of conventional econometric approaches to achieve accurate financial projections.

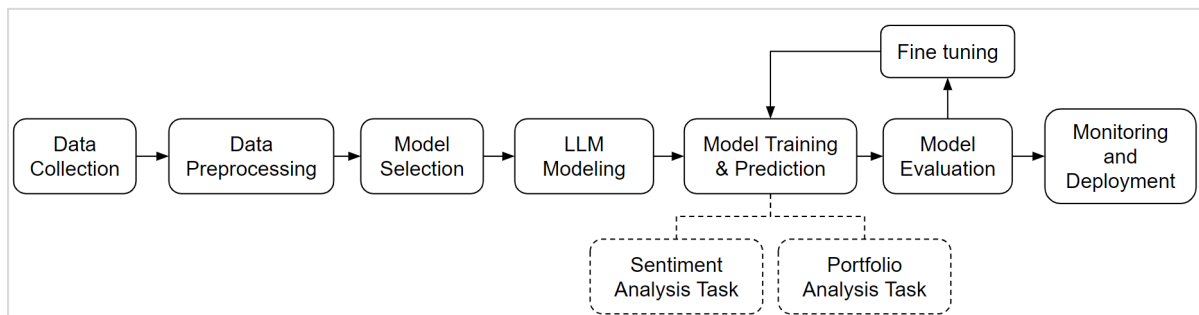


Figure 2. Methodology of LLM Application for Communication Industry Stock Forecasting

The diagram shown in Figure 2 provides the complete methodology of this study, starting by collecting data from Wharton Research Data Services (WRDS), including extensive datasets from Centre for Research in Security Prices (CRSP) for stock returns and IQ Key Development for headline news. After collecting the data, the next step is data preparation, which includes cleaning and transforming the data.

The process of model selection has significance as it involves choosing suitable LLMs and econometric models for analysis. Pre-trained models include the extraction of average embeddings and the application of logistic regression to estimate sentiment. On the other hand, fine-tuned models do sentiment analysis directly, without the need for extracting embeddings as an intermediary step.

The framework employs rolling window analysis to ensure the stability and correctness of the model over time. This systematic methodology guarantees a comprehensive analysis, using the advantages of both LLMs and conventional econometric approaches to provide precise financial forecasts and valuable insights.

3.2 Hardware Specifications

The study used computing resources offered by Google Colab Pro. This platform provides powerful hardware capabilities that facilitate the processing, analysis, and training of models required for this study. The Table 2 below provides a comprehensive breakdown of the specs for the GPU model utilised:

Table 2. Google Colab Pro L4 GPU Specification

Specification	Description
GPU Model	NVIDIA L4
GPU Memory	24 GB GDDR6
GPU Temperature	Typically between 60-80°C
GPU Power Usage (Max)	120W
GPU Memory Usage	Up to 24 GB
System RAM	52 GB
Disk Space	200 GB SSD

Note: This table outlines the hardware specifications of the Google Colab Pro L4 GPU, including GPU model, memory, operating temperature, power usage, and system specifications.

For this study, over 300 Colab units were used. The substantial use of computer resources facilitated the effective management of large datasets, training of complex models, and execution of the requisite calculations for sentiment analysis and return forecasts. The combination use of the high-performance NVIDIA L4 GPU and system RAM were crucial in training the deep learning models, resulting in a substantial reduction in processing time compared to using just the CPU.

3.3 Data Collection

The data for this research is obtained from the WRDS, which consists of the CRSP for stock returns and IQ Key Development for headline news. The dataset spans from 2005 to 2023, with data from 2005 to 2015 used for training (in-sample) and data from 2016 to 2023 used for validation (out-of-sample).

CRSP Dataset: The CRSP data offers extensive financial information, as shown in the Table 3 below:

Table 3. CRSP Dataset Variables

Field	Description
Date	The specific date of each record
Ticker	The stock ticker symbol of the company
Company Name	The name of the company
PERMCO	A unique identifier for each company
Price	The closing price of the stock
Volume	The number of shares traded
Return	The stock return, calculated as the percentage change in price
Shrout	The number of shares outstanding

***Note:** This table outlines the key variables included in the CRSP dataset, which are used for the analysis of stock market performance. Each variable provides essential information about the stock trading activities and financial metrics of the listed companies.*

The IQ Key Development dataset comprises comprehensive news and event data, as shown in the Table 4 below:

Table 4. IQ Key Development Dataset Variables

Field	Description
Company Name	The name of the company involved in the event
Event Type	The type of event (e.g., earnings announcement, merger, acquisition)
Announce Date	The date the event was announced
Headline	The headline of the news article
Situation	The full body of the headline, providing more context and details about the event
Keydevid	A unique identifier for each event

***Note:** This table presents the variables included in the IQ Key Development dataset. These variables capture essential details about significant company events, which are crucial for understanding the context and impact of news articles on stock market performance.*

For a more detailed data collection process, please refer to Appendix B.

3.4 Data Preprocessing

Data preparation involves a series of crucial tasks aimed at preparing the data for analysis. The method starts with choosing the Top-25 companies from the CRSP dataset, based on their largest market capitalization, that were listed as of the year 2005. This selection guarantees that the study concentrates on established and important companies in the US Communication Service Industry sector.

After selecting the companies, the daily returns are converted into weekly returns and are accomplished by using adjusted pricing. It is essential to accurately account for stock splits and other company activities when calculating the adjusted price of the company and

ensuring the returns in order to retain accuracy. This phase guarantees that the returns accurately represent the actual economic worth and fluctuations in stock prices.

Data normalisation is then used to standardise numerical data across disparate scales. Normalisation guarantees the analysis is not influenced by the differing scales of characteristics, enabling more significant comparisons of model training.

The subsequent step is to eliminate identical headlines that occur on the same day by retaining the latest '*situation*' variable that has the most complete information in order to prevent repetition in sentiment analysis. It is crucial to do this step in order to guarantee that each headline makes a distinct contribution to the sentiment score or probability, without adding any bias via repetition.

Dealing with absent headlines is another crucial component of preprocessing. Dates that do not have headlines are replaced with a '*[No_Headline]*' placeholder in order to ensure the consistency of the dataset. This method guarantees the systematic documentation of the lack of news, which may serve as a helpful indication in sentiment analysis.

Ultimately, the process of combining the CRSP and IQ Key Development datasets is performed in order to create a unified dataset that can be used for analysis. This ensures that financial data and headline news are appropriately synced by aligning each input based on date and its respective company.

3.5 Rolling Windows Analysis

The rolling window technique guarantees that the models are consistently updated and verified using current data, capturing any evolving trends and patterns. This methodology entails partitioning the dataset into several windows, each including a training phase and a validation period. More precisely, the study utilises 8 windows, with each period consisting of 10 years of training data and 1 year of validation data. Figure 3 below depicts the process of rolling windows analysis in this study, displaying the periods of training and validation across the whole series spanning from 2005 to 2023.

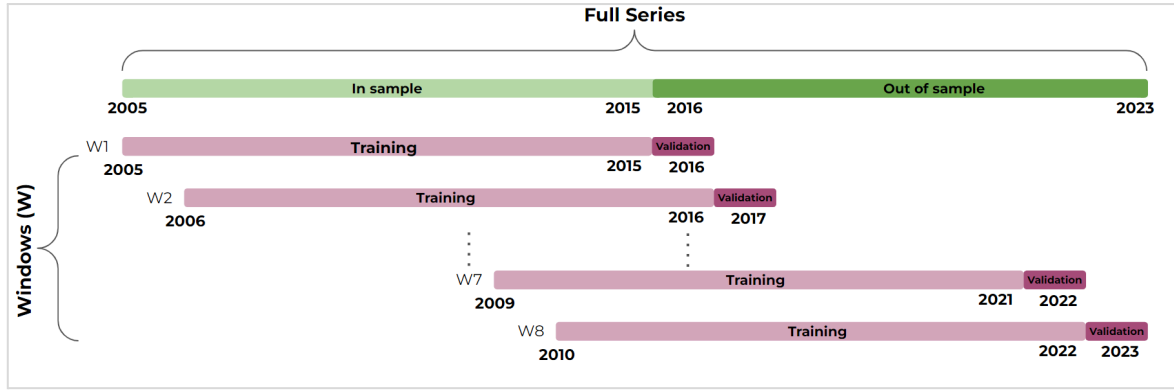


Figure 3. Rolling Windows Analysis

The approach starts by delineating each window to include a training time of 10 years, followed by a validation phase of 1 year. During the training phase, the models are trained using the data from the specified training period. Afterwards, the validation phase consists of evaluating the model's performance by utilising the data from the succeeding validation period. Following each validation period, the models are refreshed with the latest data, resulting in a one-year advancement of the time horizon.

3.6 Sentiment Analysis

3.6.1 Pre-trained

LLMs are used to do sentiment analysis on headline news, extracting the emotional tone associated with the text. At first, five pre-trained models are employed: BERT, RoBERTa, FinBERT, DistilBERT, and DistilRoBERTa.

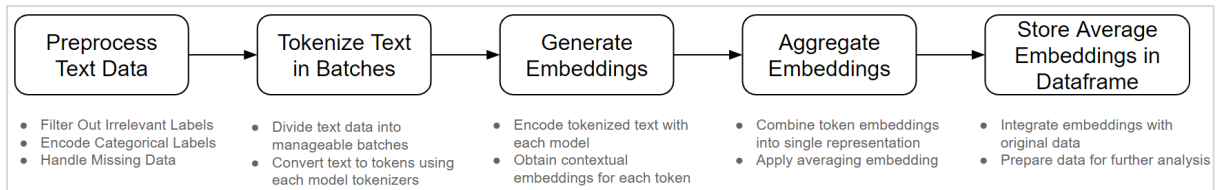


Figure 4. Average Embeddings Extraction Process

Pre-trained models start the process by obtaining average embeddings and then use logistic regression to determine sentiment probability and the price direction compared to the previous and the next week (up or down). This process involves many stages as shown in Figure 4 above. Initially, the text data undergoes preprocessing to eliminate extraneous labels, encode category categories, and address missing data. Subsequently, the text is partitioned into batches, breaking down the text data into manageable chunks and transforming it into tokens using the tokenizers specific to each model. After the text is divided into tokens, the embeddings are created by encoding the tokenized text using each model. This process results

in obtaining contextual embeddings for each individual token. The token embeddings are consolidated into a unified representation by using an averaging procedure. Ultimately, the average embeddings are saved in a dataframe, merging the embeddings with the original data and making it ready for further analysis.

3.6.2 Logistic Regression

Logistic Regression is used to create a model that predicts the likelihood of a binary event, such as whether stock price will go up or down, based on sentiment probabilities from the LLMs. The equation below describes the connection between the labels and features:

$$E(y_i, t \mid x_i, t) = \sigma(x_i, t\beta)$$

In this context, $\sigma(x)$ represents the logistic link function, which transforms the characteristics into a numerical value ranging from 0 to 1. By incorporating Logistic Regression with pre-trained models, it improves the capacity to reliably score the sentiment, which is vital for forecasting stock fluctuations.

3.6.3 Fine-tuned

Unlike the process of obtaining average embeddings and utilising logistic regression, the fine-tuned models immediately carry out sentiment analysis. The fine-tuned SequenceClassifier model predicts both the price direction and the likelihood of negative or positive emotion. This efficient method utilises the improved functionalities of finely-tuned models to provide precise sentiment forecasts.

The efficacy of the sentiment analysis is assessed by Accuracy, which guarantees the models' proficiency in capturing the sentiment expressed in the headlines. The Table 5 outlines the distinctions between pre-trained and fine-tuned models for the sentiment analysis task:

Table 5. Pre-trained (Untuned) vs Fine Tuned Models for Sentiment Analysis Task

Model	Pre-trained Models	Fine Tuned Models for Sentiment Analysis Task
BERT	'BertModel'	'BertForSequenceClassification'
RoBERTa	'RoBERTaModel'	'RobertaForSequenceClassification'
Distil BERT	'DistilBertModel'	'DistilBertForSequenceClassification'
Distil RoBERTa	'DistilRobertaModel'	'DistilRobertaForSequenceClassification'
FinBERT	'finbert-pretrain'	Finbert-tone 'BertForSequenceClassification'

Note: This table contrasts pre-trained and fine-tuned models used for sentiment analysis in this study. Pre-trained models are initially trained on general datasets, while fine-tuned models are further trained on sequence classification tasks.

3.7 Portfolio Construction & Return Prediction

The final phase is forecasting stock returns by incorporating sentiment data into econometric models. The sentiment analysis results from the LLMs are combined with cumulative log returns to improve the forecasting precision of the models. More precisely, the models are trained to predict stock returns by using cumulative log returns, which provide a more reliable measure of returns over a period of time in comparison to plain arithmetic returns. Cumulative Logarithmic (log) return is calculated using the following formula:

$$\text{Cumulative Log Return} = \sum_{t=0}^n \log(1 + \text{Weekly Return } t)$$

This measure incorporates the compounding impact of returns over many time periods, resulting in a more precise depiction of investment success over time.

A total of 12 methods are used for portfolio development, including both untuned and fine-tuned models from the five distinct LLMs. The strategies are outlined below:

- Equal Weighted Long: Creating a portfolio by allocating equal amounts of investment to equities that are expected to provide positive returns.
- Equal Weighted Short: Creating a portfolio by evenly investing in equities that are expected to have negative returns.
- The Equal Weighted Long-Short strategy involves creating a portfolio with no net investment by going long on the top 20% of companies with the most positive sentiment ratings and shorting the bottom 20% of stocks with the most negative sentiment scores.
- Value Weighted Long: Creating a portfolio by investing in companies that are expected to have good returns, with the investment amount determined by their market capitalization.
- Value Weighted Short: Creating a portfolio by investing in companies that are expected to have negative returns, with the investment amount determined by their market capitalization.
- The Value Weighted Long-Short strategy involves creating a portfolio with no net investment by going long on the top 20% of stocks with the highest positive sentiment scores and shorting the bottom 20% of stocks with the most negative sentiment scores. The weights of these positions are determined by the market capitalization of the stocks.

The portfolios' performance is assessed using Sharpe Ratios, which quantify the risk-adjusted return of the portfolio. The Sharpe Ratio (annualised) is computed using the below formula:

$$\text{Sharpe Ratio} = (R_p / \sigma_p) \times \sqrt{52}$$

where R_p is the average return of the portfolio and σ_p represents the standard deviation of the portfolio's return. This methodology guarantees that the portfolios are built with the intention of achieving the optimal performance. It utilises the predictive capabilities of LLMs and econometric methodologies to provide precise and dependable projections of stock returns.

Chapter 4: Results and Discussion

4.1 Statistical Overview

This section presents a comprehensive examination of the results of weekly returns, headline news, top tokens, the average number of tokens per headline across various models, and the visual representations of word clouds.

4.1.1 Weekly Return

The summary data for the weekly returns of companies in the US Communication Service Industry sector are extensive, including a range of parameters like mean, standard deviation, and quartiles, among others. This information is essential for understanding the performance and volatility of any company's stock. As an example, AT&T Inc. (Ticker: T) has an average weekly return of 0.001357 and a standard deviation of 0.029655. This suggests that AT&T's returns are relatively stable with moderate variability. In contrast, companies like Nexstar Media Group (NXST) have higher standard deviations (0.096741), indicating greater volatility in their stock returns. The weekly returns for AT&T range from a low of -0.189807 to a top of 0.140761. The quartiles for AT&T are as follows: Q1 is -0.014286, Q2 is 0.000891, and Q3 is 0.016607. This indicates that 25% of AT&T's weekly returns fall below -0.014286, while 75% fall below 0.016607. These insights are helpful for investors or portfolio managers who want to comprehend the risk and return characteristics of various equities. For a complete summary for each company, please refer to Appendix C.

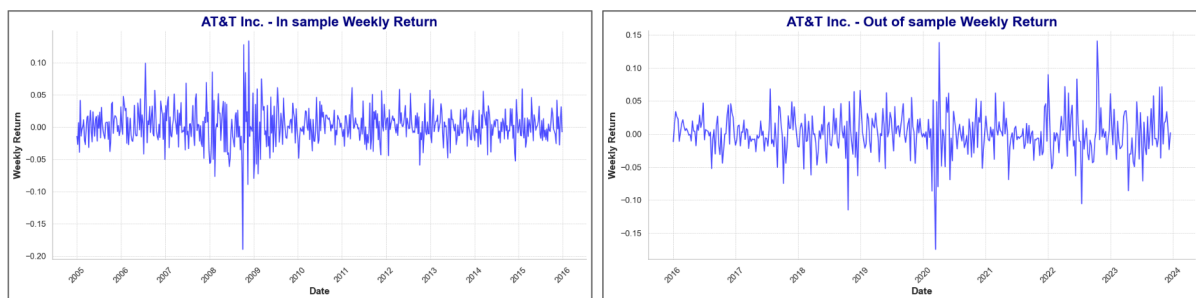


Figure 5. Weekly Return of In-sample (left) and Out-of-sample (right) for AT&T Inc.

The visualisations shown above demonstrate the weekly return for AT&T Inc. throughout the periods of analysis and validation. The chart on the left displays the returns seen during the period from 2005 to 2016. Conversely, the chart on the right illustrates the returns observed during the period from 2016 to 2024, known as the out-of-sample period. These charts provide a clear depiction of the fluctuation and patterns in AT&T's stock returns over a period

of time. To access the comprehensive in-sample and out-of-sample statistics for all companies, please refer to Appendix D.

4.1.2 Headline News

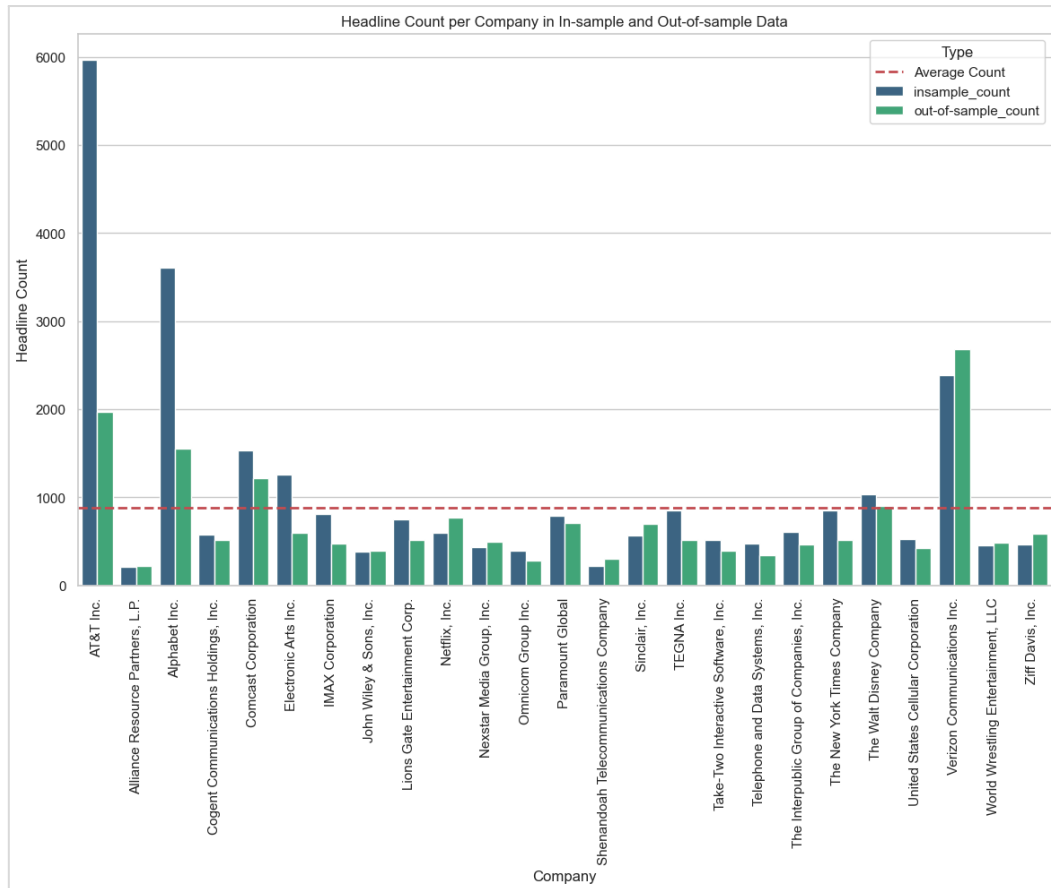
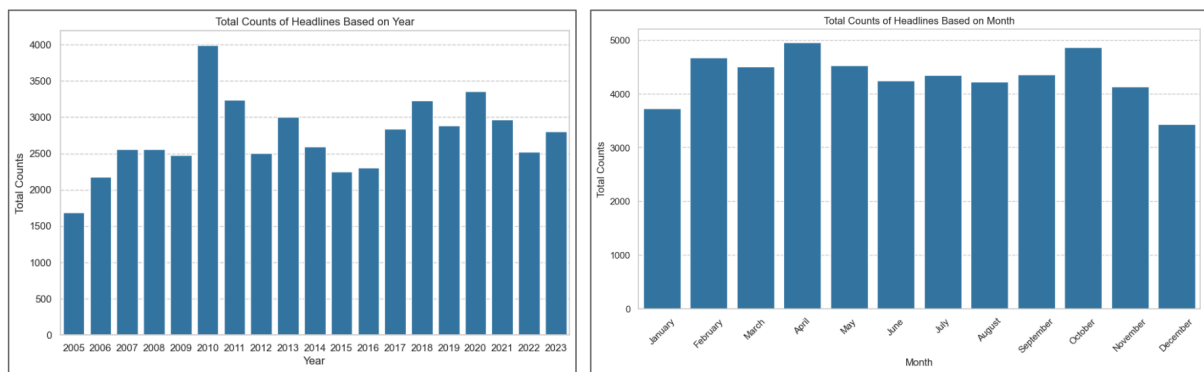


Figure 6. Headline count per-Company

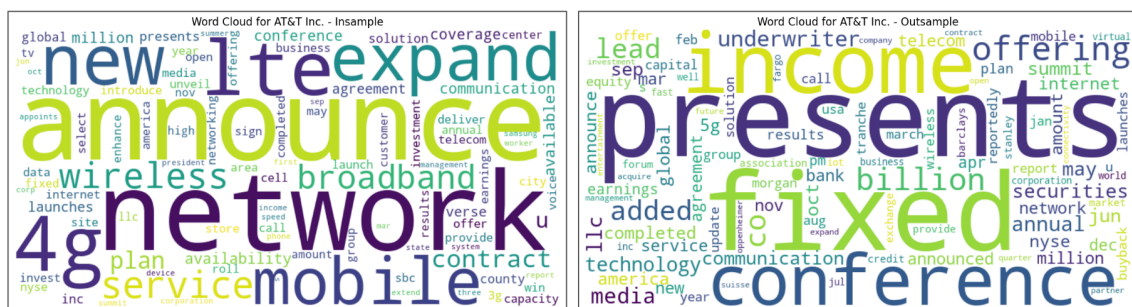
The bar chart shown in Figure 6 above displays the distribution of headline counts per company. The headline count per company offers a comprehensive perspective of the news coverage that each company gets, which is a crucial aspect in sentiment research. AT&T Inc. and Alphabet (Google) are companies that have a noticeably larger number of headlines with up to 6000 and 3500 headlines respectively, suggesting that they get news attention more often. The heightened media focus might result in higher stock price volatility when news spreads quickly and market responses occur.

The variation in the number of headlines also underscores the varying degrees of media visibility across firms. Companies with lower headline counts may encounter less market volatility as a result of a smaller number of news events that significantly impact the market. The consequences of media coverage are crucial for sentiment analysis since regular news announcements might result in more precise sentiment predictions owing to a larger dataset.



In addition, examining news headlines by year and month offers further insights regarding the temporal distribution of news coverage. The left figure in Figure 7 displays the cumulative number of headlines each year from 2005 to 2023, emphasising the yearly fluctuations in media coverage, while the figure on the right displays the overall number of headlines by month, showing a fairly uniform distribution with spikes in April and October, reflecting seasonal patterns in news coverage.

4.1.3 Word Cloud & Top Tokens



The word clouds depict the terms that appear most often in the headlines. The example used is AT&T Inc., which is divided into two datasets: in-sample and out-of-sample (*For a complete list of word clouds for all companies, please see Appendix E*). The regular occurrence of corporate announcements and earnings reports is reflected in the prevalence of tokens such as "conference," "announces," and "earnings" in financial news.

These tokens are crucial for training LLMs since they embody the fundamental topics found in the financial news corpus. The frequency of these tokens may also serve as an indicator of market sentiment trends, where certain terms may be linked to either positive or negative market movements.

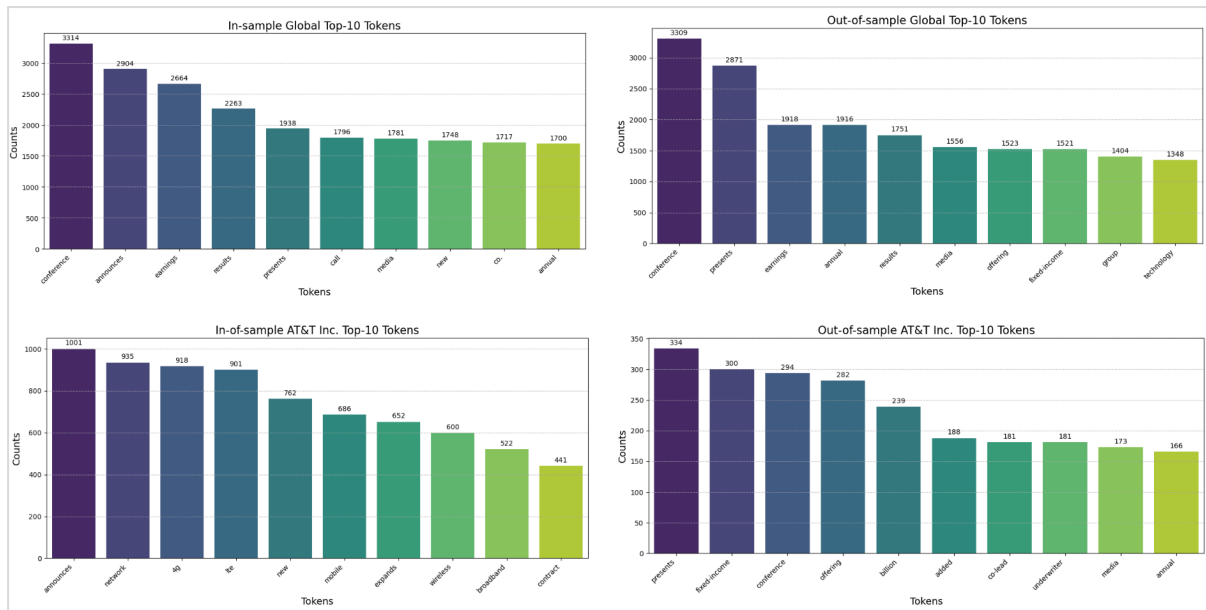


Figure 9. Top-10 Tokens in In-Sample (Top left), Out-of-Sample (Top right), and for a specific company, AT&T Inc In-sample (Bottom left) and Out-of-sample (Bottom Right)

Figure 9 displays the top-10 tokens used in headlines, both globally and particularly for AT&T Inc. as an example. The tokens are ordered based on their frequency of occurrence. During the in-sample period, the most often occurring tokens are "conference," "announces," "earnings," and "results," which suggest the prevalent subjects discussed in financial news within this specific time frame. Similarly, throughout the out-of-sample period, there is a persistent occurrence of words like "conference," "presents," "earnings," and "annual." These tokens represent distinct subjects that are pertinent to overall or on specific company sentiment's corporate operations and financial occurrences. The frequency of these tokens in the headlines indicates a concentration on presentations, debates connected to revenue, and conferences, which are crucial in the company's press coverage.

4.2 Results & Discussion

4.2.1 Model Performance

This section assesses the performance of the models using two main metrics: Accuracy for sentiment analysis in forecasting the direction of return, and Sharpe Ratios for constructing portfolios. The assessment measures used in this study include Accuracy for sentiment analysis, which gauges the models' capacity to forecast the direction of price (up or down), and Sharpe Ratios for portfolio construction.

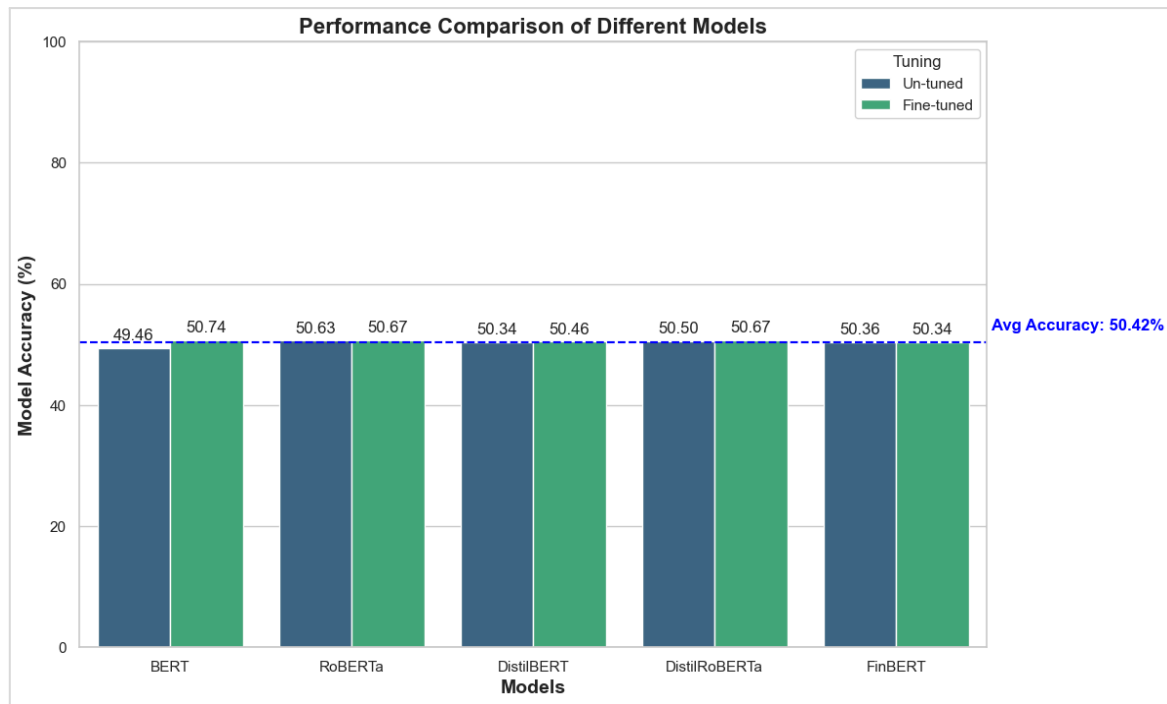


Figure 10. Model Accuracy by average Untuned vs Fine Tuned

Figure 10 presents a comparison of the predictive accuracy on the five models in determining the direction of price. Every model is assessed in both its un-tuned and fine-tuned stages and the metric demonstrates that fine-tuning typically enhances the accuracy of the models. The BERT model, after being fine-tuned, obtains the maximum accuracy of 50.74%. This indicates that it is the most successful model among the ones studied for predicting return direction. This enhancement emphasises the need of adjusting pre-existing models to suit certain financial scenarios via meticulous calibration. The mean accuracy across all models is 50.42%, suggesting a moderate but steady increase via fine-tuning. To see the complete model accuracy throughout the year, please see Appendix F.

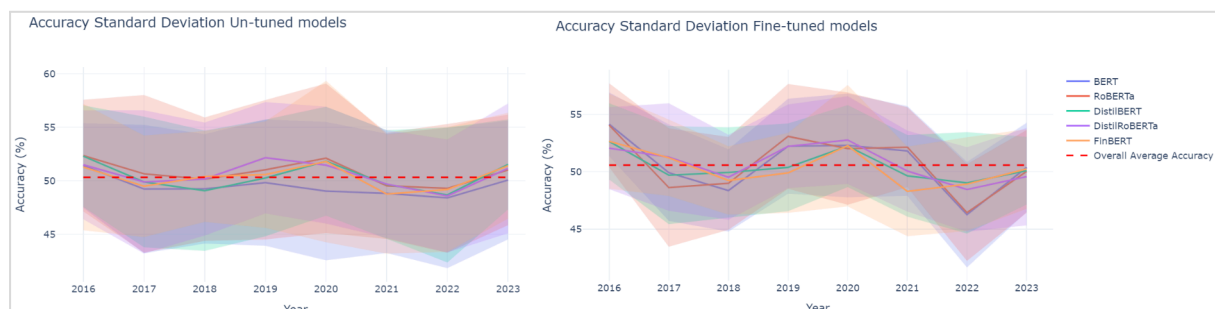


Figure 11. Models' Accuracy standard deviation

Furthermore, the standard deviation of accuracy for both un-tuned and fine-tuned models offers valuable information on the stability of the models' performance over time. The figure depicted in Figure 11 presents the standard deviation of accuracy for each model from 2016

to 2023. By fine-tuning, the standard deviation is decreased, which suggests a higher level of stability and consistency in performance. For the whole table of accuracy standard deviations, please see Appendix G.

Table 6. Sharpe Ratios of all models across different strategies

Strategy		BERT		RoBERTa		Distil BERT		Distil RoBERTa		FinBERT	
		UT	FT	UT	FT	UT	FT	UT	FT	UT	FT
EW	LS	1.11	0.95	-1.52	-1.26	1.47	1.53	-1.24	-1.28	0.73	-0.87
	L	0.20	1.39	-0.08	1.20	1.11	1.06	-0.69	-0.10	1.03	-0.28
	S	-0.67	-0.44	1.97	1.98	-0.14	-1.41	0.49	1.48	0.07	0.97
VW	LS	0.80	1.39	-2.27	-0.55	1.26	1.16	-1.35	-1.45	0.65	-1.15
	L	-0.17	2.06	0.49	0.96	0.69	0.90	-0.59	0.42	0.91	-0.85
	S	-0.74	-0.33	1.57	1.60	-0.09	-0.92	0.72	1.87	-0.23	1.23

Note: *UT = Un-tuned, FT = Fine-tuned, EW = Equal-weighted, VW = Value-weighted, LS = Long-Short, L = Long, S = Short. These terms define the different strategies for each portfolio: Long-Short (LS) involves taking long and short positions simultaneously, Long (L) refers to taking long positions only, and Short (S) refers to taking short positions only.*

The Table 6 displays the Sharpe Ratios for several portfolio strategies, including Long-Short (LS), Long (L), and Short (S), utilising both equal-weighted (EW) and value-weighted (VW) portfolios. The Sharpe Ratios demonstrate that certain models and strategies provide superior returns when adjusted for risk. The BERT model, especially when fine-tuned, consistently exhibits robust performance across different portfolio strategies. This indicates that BERT is highly suitable for financial applications, efficiently managing the trade-off between return and risk.

BERT's performance, especially in LS strategies, demonstrates its resilience and dependability in forecasting returns and handling portfolio risk. Although DistilBERT has outstanding performance, BERT is still the favoured model because of its extensive design and superior ability to capture intricate financial patterns. The findings emphasise the variation in performance across various models and tuning states, and highlights the need of carefully choosing and refining models to optimise financial forecasting models and create lucrative portfolios.

4.2.2 Return Predictions and Portfolio Performance

The results suggest that fine tuned models frequently outperform their untuned models. Specialised models such as BERT and DistilBERT offer higher accuracy and improved portfolio performance. Notably, the expected return from these models closely match the actual returns, leading to more effective investment strategies. Table 7 presents a comprehensive summary of the performance statistics for these portfolios.

Table 7. Performance of weekly news sentiment portfolios of US Communication Service Industry

	BERT						DistilBERT					
	Equal Weighted			Value Weighted			Equal Weighted			Value Weighted		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
	Ret	Std	SR	Ret	Std	SR	Ret	Std	SR	Ret	Std	SR
Ret	0.41	0.05	0.37	0.21	0.09	0.12	0.34	-0.01	0.35	0.21	0.12	0.09
Std	3.95	3.16	2.37	3.95	3.16	2.37	3.73	2.98	2.24	3.73	2.98	2.24
SR	1.39	-0.44	0.95	2.06	1.6	1.39	1.06	-1.41	1.53	0.9	-0.92	1.16

	RoBERTa						DistilRoBERTa					
	Equal Weighted			Value Weighted			Equal Weighted			Value Weighted		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
	Ret	Std	SR	Ret	Std	SR	Ret	Std	SR	Ret	Std	SR
Ret	0.37	0.01	0.3	0.22	0.12	0.1	0.35	0.01	0.3	0.22	0.12	0.1
Std	4.2	3.36	2.52	4.2	3.36	2.52	3.86	3.09	2.32	3.86	3.09	2.32
SR	1.2	1.98	-1.26	0.96	1.57	-0.55	-0.1	1.48	-1.28	0.42	1.87	-1.45

	FinBERT					
	Equal Weighted			Value Weighted		
	Long	Short	L-S	Long	Short	L-S
	Ret	Std	SR	Ret	Std	SR
Ret	0.21	0.14	0.14	0.2	0.17	0.02
Std	3.83	3.06	2.3	3.83	3.06	2.3
SR	-0.28	0.97	1.44	-0.85	1.23	-1.15

Note: The table summarises the performance metrics of different models for equal-weighted and value-weighted strategies using returns (Ret), standard deviation (Std), and Sharpe Ratios (SR) metrics.

The BERT and DistilBERT models demonstrate outstanding performance. BERT exhibits a return of 0.41 in the EW long strategy and a return of 0.21 in the VW long strategy. Additionally, the Sharpe ratios are greater in the VW strategy, indicating superior risk-adjusted returns. DistilBERT demonstrates strong performance in the EW LS strategy, achieving a return of 0.35 and a Sharpe ratio of 1.53. RoBERTa has a 0.37 return for the equally weighted (EW) long strategy and a 0.22 return for the value-weighted (VW) long strategy. It shows considerable proficiency in short positions and achieves a large Sharpe ratio of 1.87 in the VW short strategy. DistilRoBERTa adheres to a comparable pattern. FinBERT exhibits relatively poorer performance, with EW and VW long strategies yielding

returns of 0.21 and 0.20, respectively. Furthermore, its Sharpe ratios suggest a lesser ability to effectively manage risk when compared to BERT and DistilBERT.

The models used in portfolio construction will focus exclusively on the Long-Short strategy. This strategy is preferred due to its superior Sharpe ratios and its ability to minimise risk while generating higher returns. This is demonstrated by the cumulative log return of both the equal-weighted long-short (EW-LS) and value-weighted long-short (VW-LS) portfolios, for both untuned and fine-tuned models.

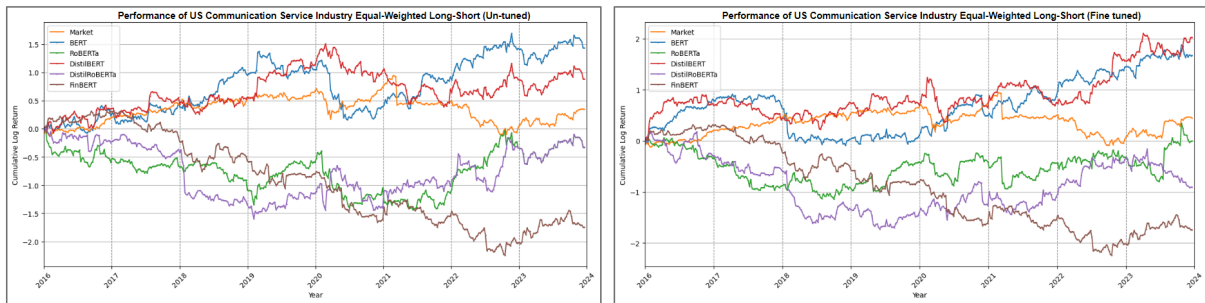


Figure 12. Equal-weighted Long-Short un-tuned (left) and Fine tuned (right) portfolios

The EW-LS method in Figure 12 is represented by two charts. The left chart displays the performance of untuned models, while the right chart illustrates the performance of models that have been fine-tuned. In general, fine-tuned models demonstrate superior performance compared to untuned models, with BERT (fine-tuned) and DistilBERT (fine-tuned) routinely outperforming other models and the industry benchmark.

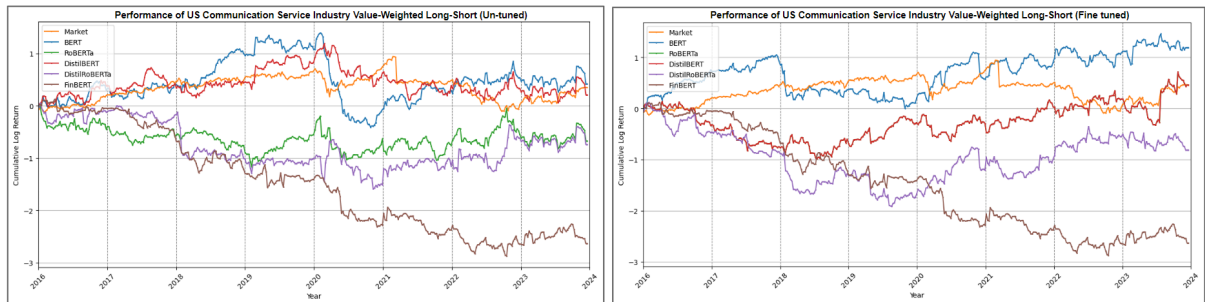


Figure 13. Value-weighted Long-Short un-tuned (left) and Fine tuned (right) portfolios

Figure 13 displays the VW-LS technique. The left chart represents the performance of models that have not been adjusted, while the right chart represents the performance of models that have been finely adjusted. Similarly, the fine-tuned models exhibit improved performance, with BERT (fine-tuned) demonstrating substantial enhancements compared to its untuned counterpart.

4.2.3 Transaction Costs Analysis

This section assesses the expenses incurred when implementing trading strategies utilising various LLM models in the US Communication Service Industry. Transaction costs are of utmost importance since they have the potential to greatly affect net profits. Table 8 presents a summary of the typical transaction costs and cost ranges for large-cap and small-cap stocks on all models.

Table 8. Transaction Cost Analysis US Communication Service Industry in basis point (bps)

Cost Category	Measure	Models				
		BERT	RoBERTa	FinBERT	Distil BERT	Distil RoBERTa
Average Cost	Large-cap	10.24	10.24	10.26	10.25	10.25
	Small-cap	20.97	21.01	21	21.02	20.98
Cost Range	Large-cap	2.28 - 19.40	0.59 - 18.85	1.50 - 19.73	1.22 - 19.33	1.55 - 19.50
	Small-cap	0.00 - 41.88	2.37 - 39.03	3.09 - 38.39	1.00 - 38.39	3.28 - 39.68

Note: The average and range of transaction costs are presented in basis points (bps). Large-cap stocks exhibit lower market impact costs compared to small-cap stocks, reflecting their higher liquidity.

The average transaction cost for large-cap stocks is roughly 10.24 bps for most models, with a slightly higher cost of 10.26 bps for FinBERT. Small-cap stocks exhibit an average transaction cost of approximately 21 bps across all models. This reflects the increased market impact costs resulting from lesser liquidity. The pricing ranges for large-cap stocks exhibit variety, with BERT reporting a range of 2.28 to 19.40 bps. This suggests that there is some fluctuation, but generally reduced costs due to increased liquidity. Small-cap equities exhibit higher volatility, with ranges reaching as high as 41.88 basis points for BERT, highlighting the increased market influence and trading challenges in smaller, less liquid companies.

The results are consistent with the findings of Frazzini et al. (2013), who observed a notable disparity in market affect costs between stocks with big market capitalization and those with little market capitalisation. The elevated expenses associated with small-cap stocks underscore the significance of taking into account liquidity and market impact while executing trading methods. Investment strategies focused on large-cap companies are expected to be more efficient in terms of cost, allowing for a higher retention of gross profits. On the other hand, strategies centred around small-cap stocks require meticulous execution to guarantee that net returns remain appealing after taking transaction costs into account.

4.2.4 Implications for Portfolio Management

The results indicate that making little adjustments greatly improves the ability of LLMs to forecast and perform well in financial applications. BERT and DistilBERT, when optimised, provide the highest level of accuracy in forecasts and demonstrate strong performance in portfolio management, making them well-suited for practical investment strategies in the US Communication Service Industry sector. Through the process of fine-tuning, models are able to enhance their ability to comprehend the subtleties of financial language and market mood, resulting in more precise forecasts. The persistent superiority of finely adjusted models compared to unadjusted models highlights the significance of customisation in financial implementations of LLMs. This customisation allows models to adjust to the individual attributes of financial data, including seasonal patterns, market fluctuations, and sector-specific news occurrences.

The efficacy of the fine-tuned models in the long-short strategies (both equal-weighted and value-weighted) demonstrates their proficiency in risk management. The fine-tuned BERT model's capability to get elevated Sharpe Ratios showcases its aptitude to provide superior risk-adjusted returns. The findings provide support for effective risk management tactics, which include maintaining a balance between long and short holdings and making accurate forecasts.

In volatile markets, it is vital for fine-tuned models to possess flexibility in order to effectively capture market emotion and trends. The cumulative log return findings indicate that the fine-tuned models have superior ability to adapt to changes in the market, constantly surpassing the performance of the market benchmark. This flexibility allows for the modification of strategy in response to market fluctuations.

Lastly, the Fine-tuned models may be used to enhance automated decision-making processes. The reliable and stable performance of the optimised BERT model may be included into automated trading systems to carry out transactions according to model forecasts, therefore simplifying the investing process and minimising human mistakes.

4.2.5 Analysis of Underperforming Models

Unlike BERT and DistilBERT that demonstrated significant improvements by fine-tuning, RoBERTa, DistilRoBERTa, and FinBERT models did not exhibit comparable performance. Multiple variables contribute to the poor performance of these models. Although RoBERTa and FinBERT exhibit robust performance in a range of natural language processing tasks, they may not adequately capture the intricate intricacies of the financial market compared to

BERT and DistilBERT. The designs of RoBERTa and FinBERT may not be well-suited for capturing the unique patterns and feelings often seen in financial news and data pertaining to the Communication Service Industry. The effectiveness of language models relies heavily on the calibre and specificity of the training data.

DistilRoBERTa and FinBERT may have encountered overfitting during the fine-tuning phase, when the models exhibited good performance on the training data but struggled to generalise to unfamiliar data. The observed outcomes may be attributed to the decreased Sharpe Ratios and cumulative log returns. The accuracy of sentiment analysis methods in assessing sentiment from financial texts might vary.

Chapter 5: Conclusion & Future Work

5.1 Conclusion

This study aims to investigate the efficacy of LLMs in forecasting stock returns and generating portfolios in the US Communication Service Industry sector. The comparative efficacy demonstrates that fine-tuned full-model LLMs, such as BERT, typically achieve better results than their distilled equivalents, but DistilBERT also exhibits significant enhancements. However, models such as RoBERTa, DistilRoBERTa, and FinBERT exhibited inferior performance, underscoring the significance of carefully choosing and optimising the model. Integrating sentiment data with stock return data improves the precision of predictive models for constructing portfolios and assessing risks, as demonstrated by greater Sharpe Ratios and better alignment between expected and actual returns. The findings emphasise the significance of customising models for financial applications, demonstrating the capability of LLMs to enhance portfolio management tactics.

5.2 Implications

This study has major implications for the management of portfolios and the analysis of financial data. Fine tuned models boost decision-making by offering more accurate forecasts of stock movements, resulting in optimum trading tactics and increased profitability. The exceptional success of the models in long-short portfolios demonstrates their ability to effectively manage risk and generate returns. Specifically, the meticulously optimised BERT model outperforms others by generating greater Sharpe Ratios and delivering superior risk-adjusted returns. Integrating these models into automated trading systems simplifies decision-making, minimises human mistakes, and enhances operational efficiency. The flexibility of well-adjusted LLMs in response to unpredictable markets provides a distinct advantage, and their use may expand beyond portfolio management to include other domains of financial research, such as credit risk evaluation and economic prediction.

5.3 Limitations

Although the findings show promise, it is important to realise numerous limitations. Although fine-tuned models have shown satisfactory performance overall, specific models like RoBERTa and FinBERT have not attained comparable levels of performance. This underscores the need for more study to enhance and optimise these models, particularly for financial applications.

5.4 Recommendations

Further studies should prioritise the exploration of sophisticated fine-tuning methods, such as using transfer learning with more extensive financial datasets, and integrating a broader array of financial data, including worldwide market data and macroeconomic variables. Utilising ensemble learning methods to merge numerous LLMs may improve the dependability and precision of predictions. Extending the study to more sectors may confirm the applicability of the results and provide insights that are special to each area. By following these recommendations, one may fully unlock the strength of finely calibrated LLMs in financial applications, hence enhancing investment strategies and financial analysis.

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Appendices

Appendix A. Top 25 US Communication Service Stocks by Market Capitalization

No	Ticker	Company Name	Market Capital (\$B)
1	GOOGL	ALPHABET INC / GOOGLE	958B
2	DIS	DISNEY WALT CO	367B
3	NFLX	NETFLIX INC	306B
4	T	AT&T INC	289B
5	CMCSA	COMCAST CORP NEW	283B
6	VZ	VERIZON COMMUNICATIONS INC	257B
7	PARA	VIACOM INC	61.1B
8	EA	ELECTRONIC ARTS INC	45.4B
9	TTWO	TAKE TWO INTERACTIVE SOFTWARE INC	27.9B
10	GCI	GANNETT CO INC	21B
11	OMC	OMNICOM GROUP INC	20.9B
12	IPG	INTERPUBLIC GROUP COS INC	15.8B
13	NYT	NEW YORK TIMES CO	9.35B
14	NXST	NEXSTAR BROADCASTING GROUP INC	7.84B
15	WWE	WORLD WRESTLING ENTMT INC	7.12B
16	ZD	J2 GLOBAL COMM INC	6.88B
17	LGF	LIONS GATE ENTERTAINMENT CORP	6.09B
18	USM	UNITED STATES CELLULAR CORP	5.71B
19	TDS	TELEPHONE & DATA SYSTEMS INC	4.48B
20	CCOI	COGENT COMMUNICATIONS GROUP INC	4.28B
21	SBGI	SINCLAIR BROADCAST GROUP INC	4.06B
22	ARLP	ALLIANCE RESOURCE PARTNERS	3.79
23	JW	WILEY JOHN & SONS INC	3.42B
24	SHEN	SHENANDOAH TELECOM CO	3.05B
25	IMAX	IMAX CORP	3.01B

Note: This table lists the top 25 companies in the US Communication Service sector, ranked by their market capitalization in billions of dollars.

Appendix B. Detailed Data Collection Process

Data Extraction from WRDS

Extracting data from the WRDS API (Wharton Research Data Services), specifically from the Capital IQ-Key development and CRSP databases. This step involved retrieving comprehensive financial and market data based on the PERMNOs and SIC of the selected stocks.

Steps for Extracting Data from CRSP Database

1. Define Date Range and SIC Codes:
 - Date Range: Specify the industry (SIC codes) and the date range for the estimation period.
 - SIC Code: 4899 (Communication Services)
2. Join crsp.dsf with crsp.dsenames to fetch comprehensive stock data, including company names and tickers.
3. Data Points Extracted:
 - Fields: CUSIP, PERMNO, PERMCO, company name, ticker, industry code, date, price, volume, share adjustment factor, shares outstanding, returns with and without dividends.

Steps for Extracting Data from IQ Key Developments Database

1. Define Entities (GVKEYs):
 - Entities: Specify the entities (GVKEYs) for which key development data is needed.
2. Date Range:
 - Announce Date: Set the date range for filtering relevant events.
3. Retrieve all fields from ciq_keydev.wrds_keydev where GVKEY matches the specified list and announce date falls within the defined estimation period.
4. Data Points Extracted:
 - Fields: Company Name, Event Type, Announce Date, Headline, Situation, Keydevid.

Merging CRSP and IQ Key Developments Data

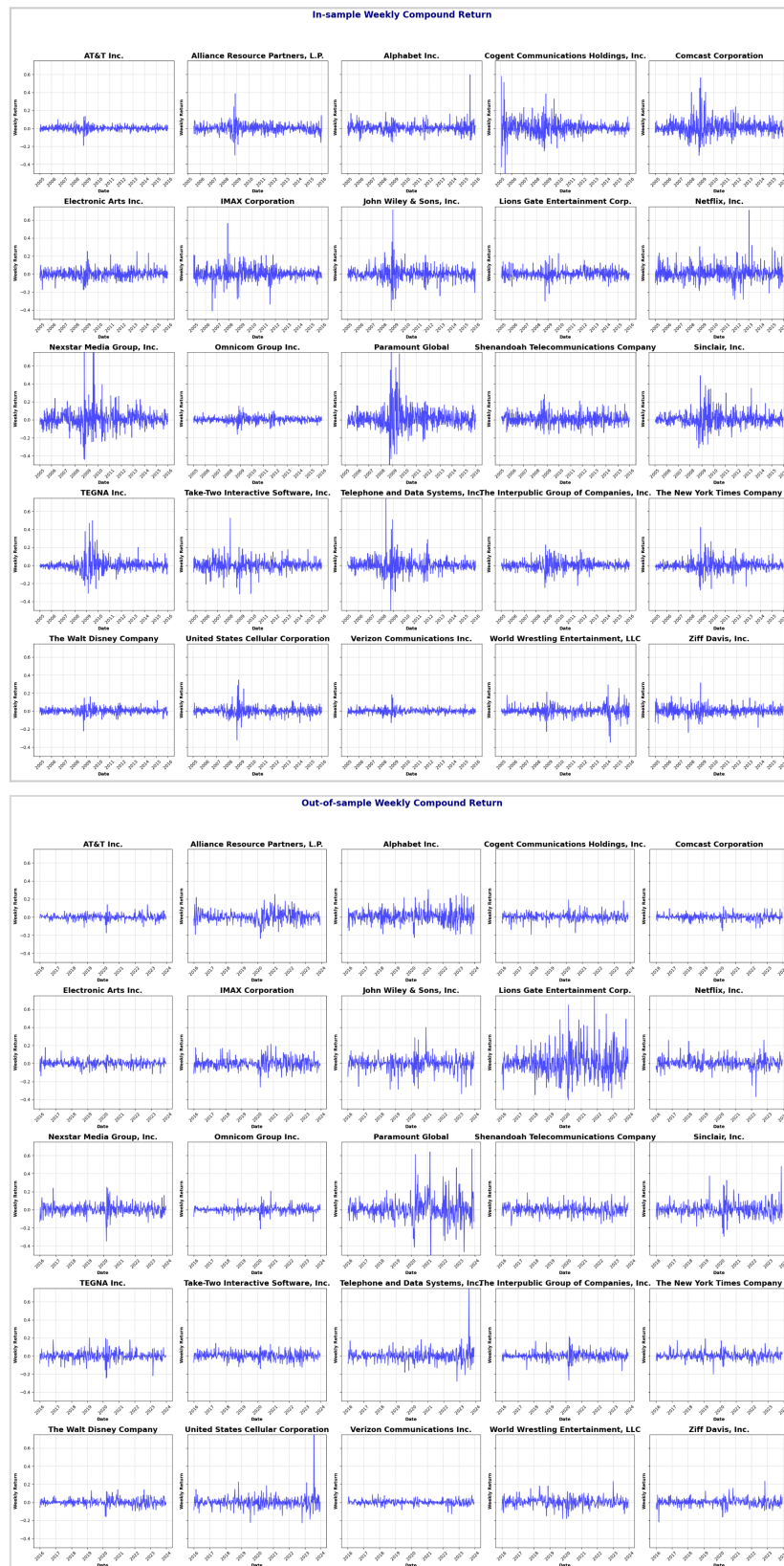
1. Join Tables:
 - Merge the CRSP data with the IQ Key Developments data based on common identifiers and date ranges.
2. Final Data Points:
 - Fields: Date, Ticker, Company Name, PERMNO, PERMCO, Price, Volume, Return, Shroud, Headline, Event Type, Situation, Keydevid.

Appendix C. Summary statistics of weekly return all company

No	Ticker	count	mean	std	min	q1	q2	q3	max
1	GOOGL	991	0.006956	0.063279	-0.226603	-0.027436	0.007135	0.036506	0.596943
2	DIS	991	0.002038	0.03559	-0.220041	-0.01685	0.000651	0.019732	0.160438
3	NFLX	991	0.008233	0.072586	-0.368218	-0.029329	0.004971	0.041429	0.70979
4	T	991	0.001357	0.029655	-0.189807	-0.014286	0.000891	0.016607	0.140761
5	CMCSA	991	0.004918	0.066175	-0.300594	-0.026259	0.002003	0.031566	0.564535
6	VZ	991	0.001347	0.027135	-0.128871	-0.014429	0.001186	0.01634	0.183014
7	PARA	991	0.007398	0.121324	-0.747803	-0.049161	0.004355	0.055571	0.779581
8	EA	991	0.001876	0.046061	-0.175372	-0.023307	0.002392	0.02697	0.252592
9	TTWO	991	0.003558	0.056641	-0.317409	-0.025348	0.003366	0.032079	0.526498
10	GCI	991	0.001405	0.06221	-0.309527	-0.023766	0.000359	0.024484	0.498432
11	OMC	991	0.001803	0.034112	-0.213004	-0.015565	0.002005	0.020146	0.206783
12	IPG	991	0.002518	0.049037	-0.268264	-0.02094	0.000836	0.026805	0.229858
13	NYT	991	0.001877	0.054126	-0.27248	-0.022565	-0.000609	0.023862	0.423671
14	NXST	991	0.007267	0.096741	-0.443716	-0.032464	0.004848	0.042007	1.132812
15	WWE	991	0.004104	0.051425	-0.347046	-0.02205	0.003174	0.030026	0.293192
16	ZD	991	0.00278	0.045947	-0.236577	-0.021563	0.001912	0.026021	0.315364
17	LGF	991	0.004318	0.103868	-0.406322	-0.037402	0.001351	0.039848	0.781998
18	USM	991	0.001594	0.057517	-0.320665	-0.022969	0.000728	0.024425	0.888143
19	TDS	991	0.002339	0.076904	-0.524436	-0.030282	-0.001001	0.0336	0.848071
20	CCOI	991	0.004506	0.071181	-0.700179	-0.024925	0.00341	0.035339	0.578946
21	ARLP	991	0.004199	0.078264	-0.313432	-0.037042	0.000927	0.037297	0.490566
22	SBGI	991	0.003083	0.054803	-0.297794	-0.025542	0.001846	0.031268	0.386911
23	JW	991	0.001896	0.075018	-0.405347	-0.03004	0.002101	0.036611	0.715546
24	SHEN	991	0.003824	0.057182	-0.177703	-0.032294	0.003415	0.036791	0.281814
25	IMAX	991	0.002876	0.067145	-0.408602	-0.032889	0.002567	0.035972	0.564103

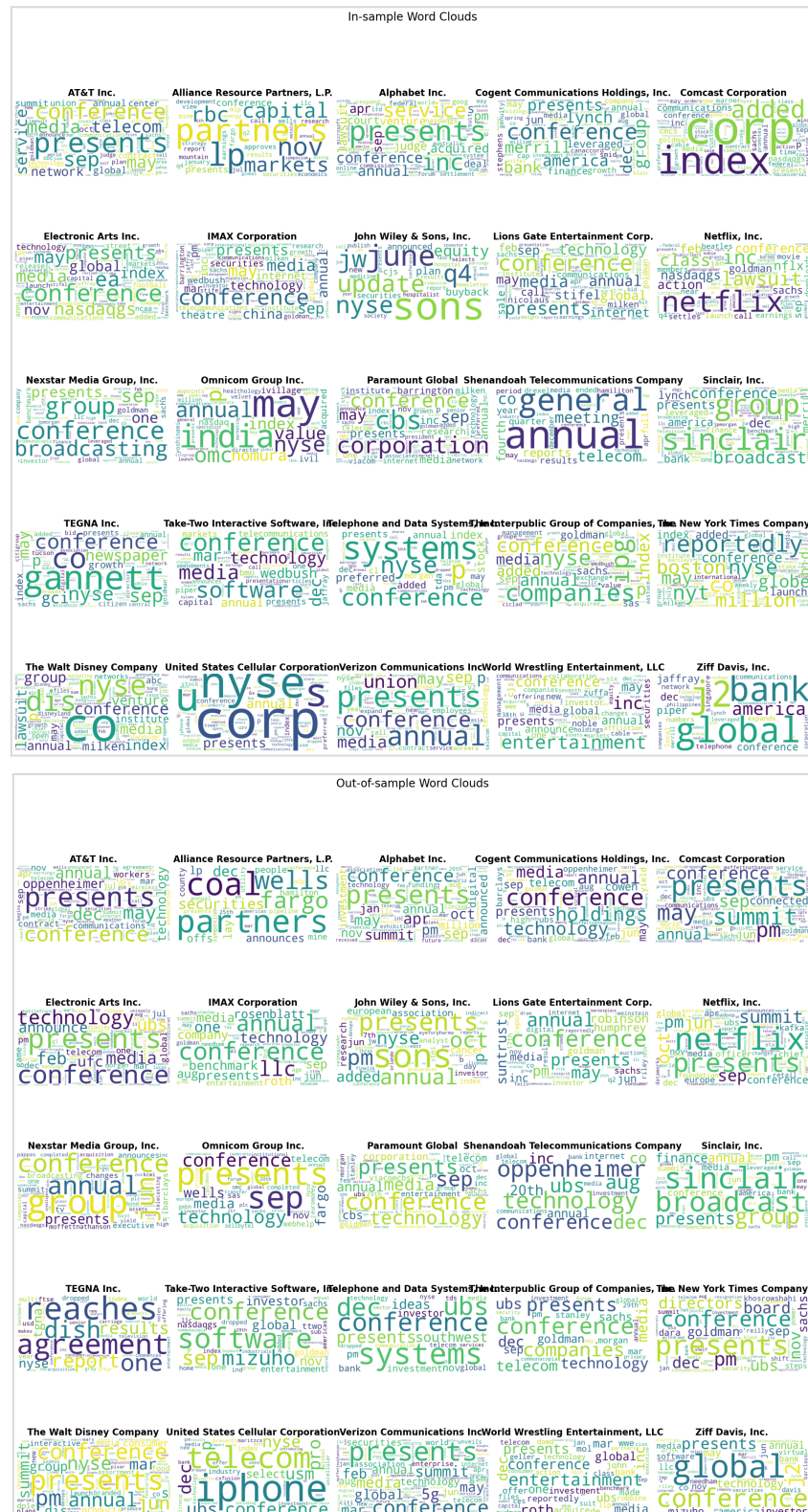
Note: This table presents the summary statistics of weekly returns for all companies in the study. The columns include the ticker symbol, count of observations, mean return, standard deviation (Std Dev), minimum (Min), first quartile (Q1), median (Q2), third quartile (Q3), and maximum (Max) weekly returns for each company. This provides a comprehensive view of the return distribution for each stock.

Appendix D. Weekly Return each company in-sample (top) and out-of-sample (bottom) datasets



Note: These visualisations depict the weekly returns for each company in the US Communication Service Industry sector. The top row represents the in-sample dataset, while the bottom row shows the out-of-sample dataset. Each chart illustrates the volatility and return trends over the respective periods, providing a comprehensive view of the stock performance for each company.

Appendix E. Word cloud each company in-sample (top) and out-of-sample (bottom) datasets



Note: These word clouds illustrate the most frequently occurring tokens in the headline news for each company in the US Communication Service Industry sector. The top row represents the in-sample dataset, while the bottom row shows the out-of-sample dataset. The size of each word indicates its frequency, providing insights into common themes and topics discussed in the headlines for each company.

Appendix F. Models accuracy throughout the year

Year	BERT		RoBERTa		Distil BERT		Distil RoBERTa		FinBERT	
	UT	FT	UT	FT	UT	FT	UT	FT	UT	FT
2016	51.4	54.13	52.3	54.05	52.2	52.64	51.4	52.05	51.2	52.65
2017	49.2	49.92	50.6	48.62	49.8	49.7	49.8	51.27	49.4	51.21
2018	49.2	48.36	50.1	48.99	49	49.94	50.1	49.52	50.3	49.24
2019	49.8	52.22	51	53.08	50.2	50.38	52.1	52.21	50.5	49.9
2020	49	52.29	52.1	52.01	51.8	52.23	51.4	52.76	51.7	52.27
2021	48.8	51.81	49.5	52.14	49.6	49.64	49.6	50.06	48.7	48.29
2022	48.3	46.26	49.2	46.44	48.6	49.03	48.5	48.45	49.7	48.92
2023	50	50.38	51	50.03	51.5	50.09	51.1	49.57	51.4	50.24
Avg	49.46	50.74	50.63	50.67	50.34	50.46	50.50	50.67	50.36	50.34

Note: The table above presents the accuracy of various models in their untuned (UT) and fine-tuned (FT) from 2016 to 2023. This data illustrates how fine-tuning generally improves the accuracy of the models over time.

Appendix G. Models' standard deviation throughout the year

Year	BERT		RoBERTa		Distil BERT		Distil RoBERTa		FinBERT	
	UT	FT	UT	FT	UT	FT	UT	FT	UT	FT
2016	3.95	2.77	5.21	3.65	4.74	3.32	5.09	3.56	5.9	4.13
2017	6.0	4.2	7.36	5.15	6.1	4.27	6.69	4.68	4.75	3.33
2018	5.11	3.58	5.77	4.04	5.62	3.93	5.28	3.7	4.25	2.98
2019	5.9	4.13	6.51	4.56	5.45	3.82	5.2	3.64	4.97	3.48
2020	6.46	4.52	6.98	4.89	5.09	3.56	5.46	3.82	7.54	5.28
2021	5.57	3.9	4.96	3.47	5.07	3.55	5.03	3.52	5.57	3.9
2022	6.55	4.59	6.01	4.21	6.31	4.42	5.27	3.69	5.83	4.08
2023	5.54	3.88	5.15	3.61	4.2	2.94	6.06	4.24	4.95	3.47
Avg	5.64	3.95	5.99	4.20	5.32	3.73	5.51	3.86	5.47	3.83

Note: The table above details the standard deviation of accuracy for different models in their untuned (UT) and fine-tuned (FT) states from 2016 to 2023. This data provides insight into the consistency of model performance, highlighting the reduced variance in accuracy achieved through fine-tuning.

Appendix H. Python (Code) Documentation

Documentation	Github Repository Link
Data Preprocessing & EDA	Link
Embeddings	Link
Pre-trained	Link
Fine-tuned	Link